

[Chapter 8](#) explores dataset engineering techniques in detail, including data quality evaluation and data synthesis.

Given the dominance of the transformer architecture, it might seem that model architecture is less of a choice. You might be wondering, what makes the transformer architecture so special that it continues to dominate? How long until another architecture takes over, and what might this new architecture look like? This chapter will address all of these questions. Whenever a new model is released, one of the first things people want to know is its size. This chapter will also explore how a model developer might determine the appropriate size for their model.

As mentioned in [Chapter 1](#), a model's training process is often divided into pre-training and post-training. Pre-training makes a model capable, but not necessarily safe or easy to use. This is where post-training comes in. The goal of post-training is to align the model with human preferences. But what exactly is *human preference*? How can it be represented in a way that a model can learn? The way a model developer aligns their model has a significant impact on the model's usability, and will be discussed in this chapter.

While most people understand the impact of training on a model's performance, the impact of *sampling* is often overlooked. Sampling is how a model chooses an output from all possible options. It is perhaps one of the most underrated concepts in AI. Not only does sampling explain many

seemingly baffling AI behaviors, including hallucinations and inconsistencies, but choosing the right sampling strategy can also significantly boost a model's performance with relatively little effort. For this reason, sampling is the section that I was the most excited to write about in this chapter.

Concepts covered in this chapter are fundamental for understanding the rest of the book. However, because these concepts are fundamental, you might already be familiar with them. Feel free to skip any concept that you're confident about. If you encounter a confusing concept later on, you can revisit this chapter.

Training Data

An AI model is only as good as the data it was trained on. If there's no Vietnamese in the training data, the model won't be able to translate from English into Vietnamese. Similarly, if an image classification model sees only animals in its training set, it won't perform well on photos of plants.

If you want a model to improve on a certain task, you might want to include more data for that task in the training data. However, collecting sufficient data for training a large model isn't easy, and it can be expensive. Model developers often have to rely on available data, even if this data doesn't exactly meet their needs.

For example, a common source for training data is [Common Crawl](#), created by a nonprofit organization that sporadically crawls websites on the internet. In 2022 and 2023, this organization crawled approximately 2–3 billion web pages each month. Google provides a clean subset of Common Crawl called the [Colossal Clean Crawled Corpus](#), or C4 for short.

The data quality of Common Crawl, and C4 to a certain extent, is questionable—think clickbait, misinformation, propaganda, conspiracy theories, racism, misogyny, and every sketchy website you’ve ever seen or avoided on the internet. A [study by the Washington Post](#) shows that the 1,000 most common websites in the dataset include several media outlets that rank low on [NewsGuard’s scale for trustworthiness](#). In lay terms, Common Crawl contains plenty of fake news.

Yet, simply because Common Crawl is available, variations of it are used in most foundation models that disclose their training data sources, including OpenAI’s GPT-3 and Google’s Gemini. I suspect that Common Crawl is also used in models that don’t disclose their training data. To avoid scrutiny from both the public and competitors, many companies have stopped disclosing this information.

Some teams use heuristics to filter out low-quality data from the internet. For example, OpenAI used only the Reddit links that received at least three upvotes to train [GPT-2](#). While this does help screen out links that nobody cares about, Reddit isn’t exactly the pinnacle of propriety and good taste.

The “use what we have, not what we want” approach may lead to models that perform well on tasks present in the training data but not necessarily on the tasks you care about. To address this issue, it’s crucial to curate datasets that align with your specific needs. This section focuses on curating data for specific *languages* and *domains*, providing a broad yet specialized foundation for applications within those areas. [Chapter 8](#) explores data strategies for models tailored to highly specific tasks.

While language- and domain-specific foundation models can be trained from scratch, it’s also common to finetune them on top of general-purpose models.

Some might wonder, why not just train a model on all data available, both general data and specialized data, so that the model can do everything? This is what many people do. However, training on more data often requires more compute resources and doesn’t always lead to better performance. For example, a model trained with a smaller amount of high-quality data might outperform a model trained with a large amount of low-quality data. Using 7B tokens of high-quality coding data, [Gunasekar et al. \(2023\)](#) were able to train a 1.3B-parameter model that outperforms much larger models on several important coding benchmarks. The impact of data quality is discussed more in [Chapter 8](#).

Multilingual Models

English dominates the internet. An analysis of the Common Crawl dataset shows that English accounts for almost half of the data (45.88%), making it eight times more prevalent than the second-most common language, Russian (5.97%) ([Lai et al., 2023](#)). See [Table 2-1](#) for a list of languages with at least 1% in Common Crawl. Languages with limited availability as training data—typically languages not included in this list—are considered *low-resource*.

Table 2-1. The most common languages in Common Crawl, a popular dataset for training LLMs. Sou (2023).

Language	Code	Pop.	CC size	
		(M)	(%)	Cat.
English	en	1,452	45.8786	H
Russian	ru	258	5.9692	H
German	de	134	5.8811	H
Chinese	zh	1,118	4.8747	H
Japanese	jp	125	4.7884	H
French	fr	274	4.7254	H
Spanish	es	548	4.4690	H
Italian	it	68	2.5712	H
Dutch	nl	30	2.0585	H
Polish	pl	45	1.6636	H
Portuguese	pt	257	1.1505	H
Vietnamese	vi	85	1.0299	H